Chapter 2 · Moving Objects Have Energy

Teacher Key · Guided Notes

Name: _____ **Homeroom:** ○ 4A ○ 4B ○ 4C ○ 4D

KEY *Bold slate = exact answer · Italic gray = sample / acceptable response.*

BQ How are energy, change, and movement of objects related?

My first thinking

Sample: "A moving marble makes another marble move." (Accept any moving-object-causes-change response.)

V1 VOCABULARY – MOTION

Reader p. 7

motion – the process of an object changing **position** .

N1 MOVING OBJECTS CAN CAUSE CHANGES

Reader p. 8

A moving object can cause a change .

Players cause their own motion. A ball does not. Something else puts the ball into motion.

Some objects cause their own motion. Other objects are put into motion by **outside** motion.

V2 VOCABULARY – ENERGY OF MOTION

Reader p. 9

energy of motion – the energy an object possesses while it is **moving** .

N2 MOVING OBJECTS HAVE ENERGY

Reader p. 9

All moving objects have energy of motion .

Energy is the ability to cause change. A moving object causes change, so it has energy.

MOVING OBJECT → **CHANGE** → **EVIDENCE**

> **OBSERVE** Marble collision – one moving marble rolls into a target marble.

BEFORE	AFTER
Target marble is still. *The other marble rolls toward it.*	*The target marble is moving.* *It changed from still to moving.*

Accept any sketch that shows the target still, then moving. The change matters more than the drawing.

What changed?

Sample: "The target marble changed from sitting still to moving." (Accept any before/after change.)

N3 PEOPLE USE THE ENERGY OF MOTION

Reader pp. 10–12

Write or draw the change each moving object causes.

MOVING OBJECT	WHAT CHANGE DOES IT CAUSE?
hammer (swinging)	*drives a nail into the wood*
pendulum (swinging)	*turns the clock gears*
wind (moving air)	*spins the turbine blades*

CHECK YOUR THINKING

How do we know a moving object has energy?

Start with: "We know because the moving object causes …"

Sample: "We know because it causes a change." Accept any change-as-evidence response.

BQ · NOW I THINK How are energy, change, and movement of objects related?

Sample: "A moving object has energy, and that energy causes a change." Accept any link among energy, change, and motion.